

The Classroom Engagement Inventory

Wang, Z., Bergin, C., & Bergin, D. A. (2014). Measuring engagement in fourth to twelfth grade classrooms: The Classroom Engagement Inventory. *School Psychology Quarterly*, 29(4), 517-535.
<https://doi.org/10.1037/spq0000050>

		Behavioral - Effortful			
	Affective	Behavioral-Compliance	Class Participation	Cognitive	Disengagement
I feel happy.	.78				
I feel excited.	.86				
I feel interested.	.85				
I feel proud.	.84				
I feel amused (smile, laugh, have fun).	.72				
I listen very carefully.		.87			
I pay attention to the things I am supposed to remember.		.82			
I complete my assignments.		.58			
I form new questions in my mind as I join in class activities.			.68		
I actively participate in class discussions.			.70		
I work with other students and we learn from each other.			.62		
I do not want to stop working at the end of class.			.69		
I get really involved in class activities.			.84		
I think deeply when I take quizzes in this class.				.75	
I judge the quality of my ideas or work during class activities.				.51	
I search for information from difference places and think about how to put it together.				.69	
If I make a mistake, I try to figure out where I went wrong.				.76	
If I am not sure about things, I check my book or use other materials like charts.				.68	
I go back over things I don't understand.				.80	
I try to figure out the hard parts on my own.				.52	
I ask myself some questions as I go along to make sure the work makes sense to me.				.76	
I am "zoned out", not really thinking or doing class work.					.81
I let my mind wander.					.75
I just pretend like I'm working.					.79

		Behavioral - Effortful			
Factor Correlations	Affective	Behavioral-Compliance	Class Participation	Cognitive	Disengagement
Behavioral-Compliance	.64				
Behavioral - Effortful Class Participation	.89	.73			
Cognitive	.69	.75	.81		
Disengagement	.54	.75	.60	.65	

A measure of classroom engagement should have the following attributes:

1. measure all three dimensions of engagement (affective, behavioral, and cognitive)
2. be class-specific
3. use self-report
4. use individual-level items
5. be readily administered in classrooms
6. span grades 4 to 12 to allow tracking of engagement over time
7. be short enough to be added to other assessments, but long enough to be reliable
8. demonstrate measurement invariance

The Classroom Engagement Inventory

We reviewed articles purporting to measure academic engagement

- Resulted in 35 items.
- Items were adapted to be class-specific

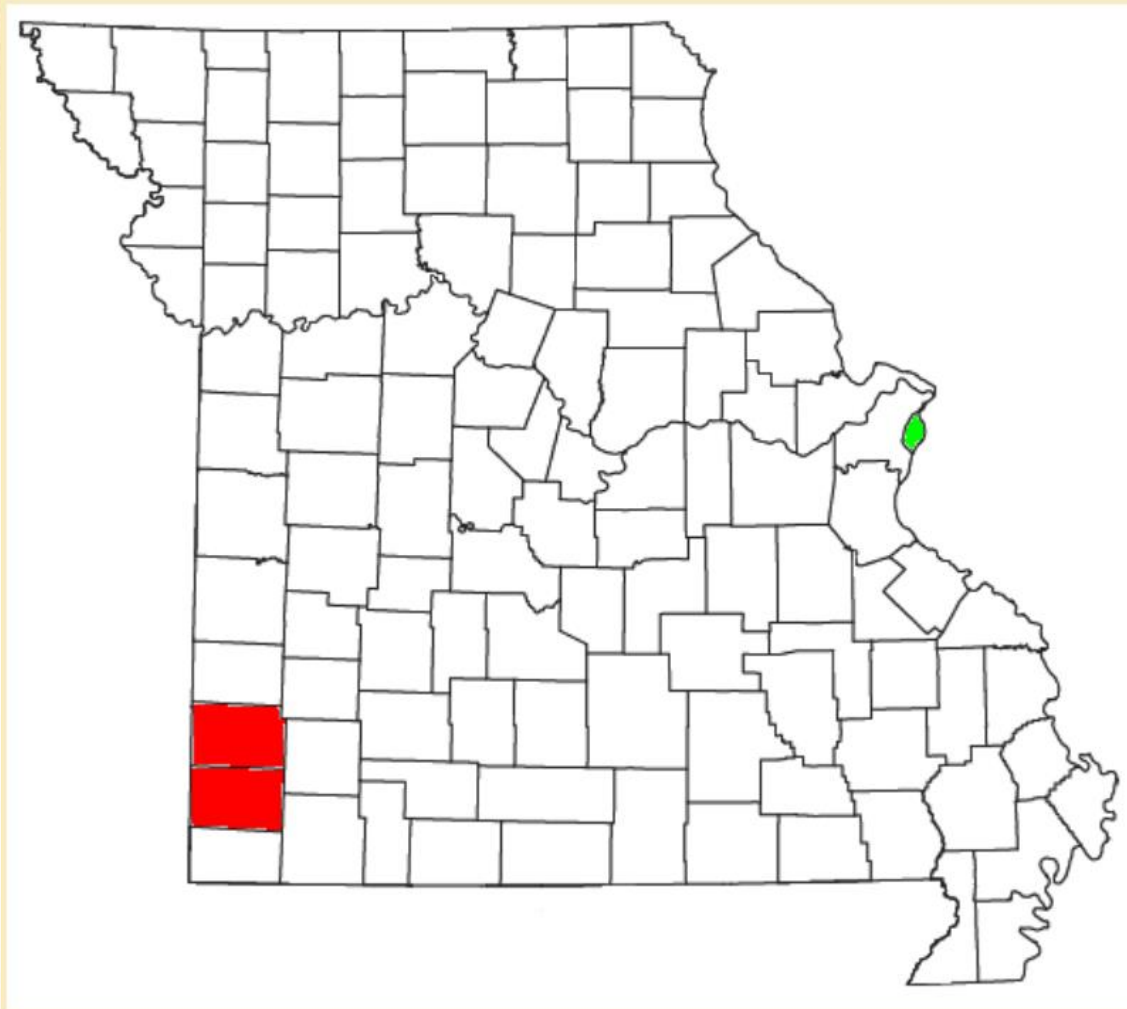


Study 1

- 3,925 students from 4th to 12th grade
- 13 elementary (K-6), 3 middle schools (6-8), 1 high school (9-12), 1 career center
- Factor analyses were based on polychoric correlations for ordered categorical variables.
- The robust mean- and variance-adjusted weighted least squares (WLSMV) estimator was used.
- A range of factor models was determined based on multiple criteria (i.e., model fit indices, modification indices, residual correlations, factor loadings, number of indicators per factor).
- EFA with Geomin rotation.
- Then CFA. During this process items that loaded less than .50 or cross-loaded on multiple factors were removed.

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Study 1



The Classroom Engagement Inventory

Four Factors Emerged

Item	Item Parameter Estimate			
	Affective	Behavioral	Cognitive	Disengaged
In this class I feel excited	0.849			
In this class I feel interested	0.849			
In this class I feel happy	0.803 ^a			
In this class I have fun	0.797			
In this class I feel proud	0.793			
I get really involved in class activities		0.822		
I actively participate in class discussions		0.776		
I form new questions in my mind as I join in class activities		0.714		
I compare things I am learning with things I already knew		0.641 ^a		
I work with other students and we learn from each other		0.625		
If I make a mistake, I try to figure out where I went wrong			0.798	
I go back over things I don't understand			0.778	
I think deeply when I take quizzes in this class			0.749 ^a	
I ask myself questions as I go along to make sure the work make sense to me			0.757	
I search for information from different places and think about how to put it together			0.727	
If I'm not sure about things, I check my book or use other materials like charts			0.671	
I judge the quality of my ideas or work during class activities			0.645	
I try to figure out the hard parts on my own			0.562	
I am "zoned out," not really thinking or doing class stuff				0.874 ^a
I let my mind wander				0.816
I just pretend like I'm working.				0.750

Note. Composite reliabilities were .91 (Affective Engagement), .84 (Behavioral Engagement), .89 (Cognitive Engagement), and .86 (Disengagement). ^a Items had unstandardized loadings fixed to unity.

Study 2

- 3,560 4th – 12th grade students in the same school district
- Revised a few items
- Scrambled items on the survey
- Added validity measures
 - Teacher practices
 - Self-Efficacy
 - Achievement Goals
 - School-Prompted Interest
 - Self-reported grades

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Study 2



Study 2

- **Step 1: Test the 4-factor CFA model derived from Study 1.** Data analysis started with the 4-factor CFA model with 21 items, derived in Study 1
 - Dropped an item
- **Step 2: Justify deletion of items and consider adding items and/or factors.**
 - Tested 7 different models.
 - The final model was a 5-factor CFA
- **Step 3: Test measurement invariance and compare latent means of groups.**
- **Step 4: Test internal consistency of the Classroom Engagement Inventory**
- **Step 5: Collect validity evidence for the Classroom Engagement Inventory**

Study 2

- **Five Factors**

- Affective Engagement
- Behavioral Engagement: Compliance
- Behavioral Engagement: Effortful Class Participation
- Cognitive Engagement
- Disengagement

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Study 2

Validity Construct	Affective	Behavioral Compliance	Behavioral - Effortful Class Participation	Cognitive	Disengagement
Teacher Behavior – Whole Class Activities	.500	.489	.571	.567	-.338
Teacher Behavior – Questioning Practices	.365	.350	.435	.408	-.168
Self-Efficacy	.553	.626	.614	.666	-.494
Mastery Goals	.600	.700	.682	.768	-.592
Performance-Approach Goals	.386	.397	.438	.488	-.305
Performance-Avoid Goals	.250	.309	.310	.385	-.219
School-prompted Interest	.566	.490	.612	.550	-.390
Self-reported Grades	.336	.387	.348	.301	-.367

Implications

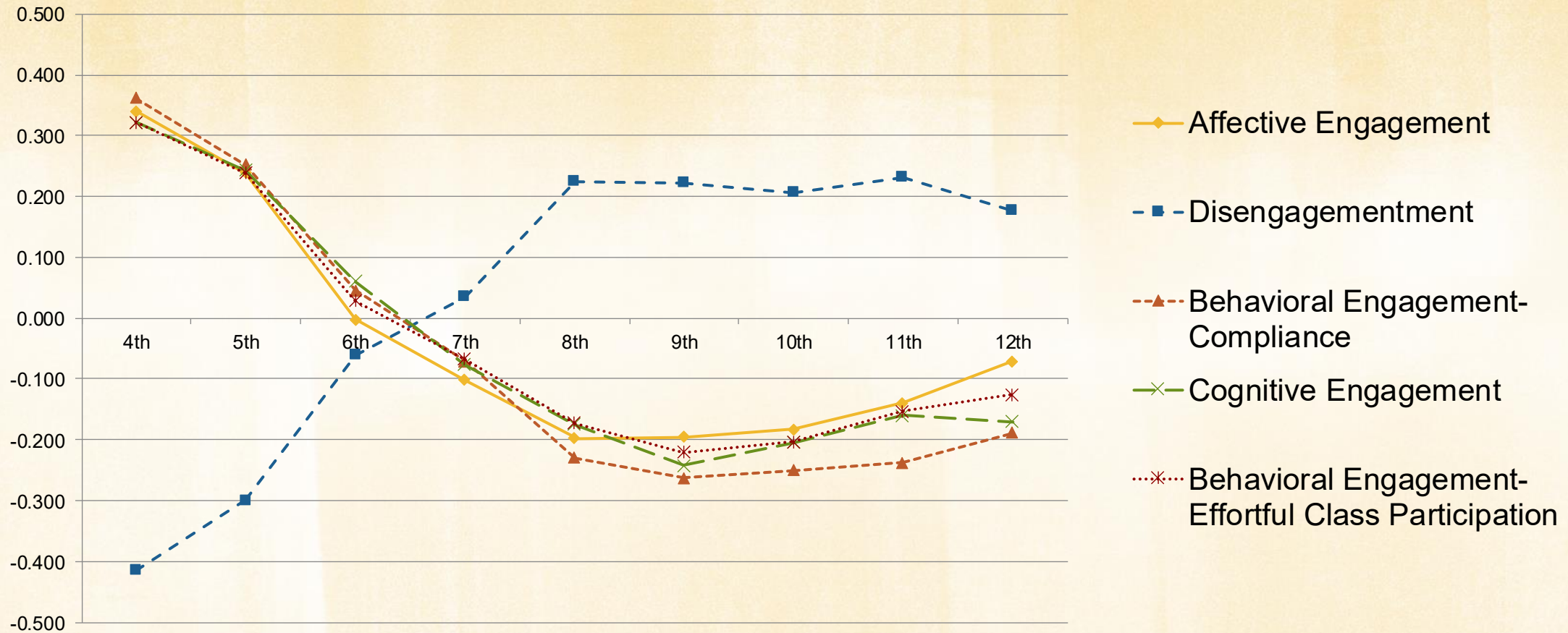
- Some have wondered if cognitive and behavioral engagement are distinct factors. The present results indicate that the CEI distinctly measures these two dimensions of engagement.
- That is, a student could be behaviorally engaged without being cognitively engaged.
- Factor structure was invariant across groups

Age Trends in Classroom Engagement

- Previous research shows a decline of motivational variables such as engagement from elementary school to high school.
- However, few studies have included later high school years.

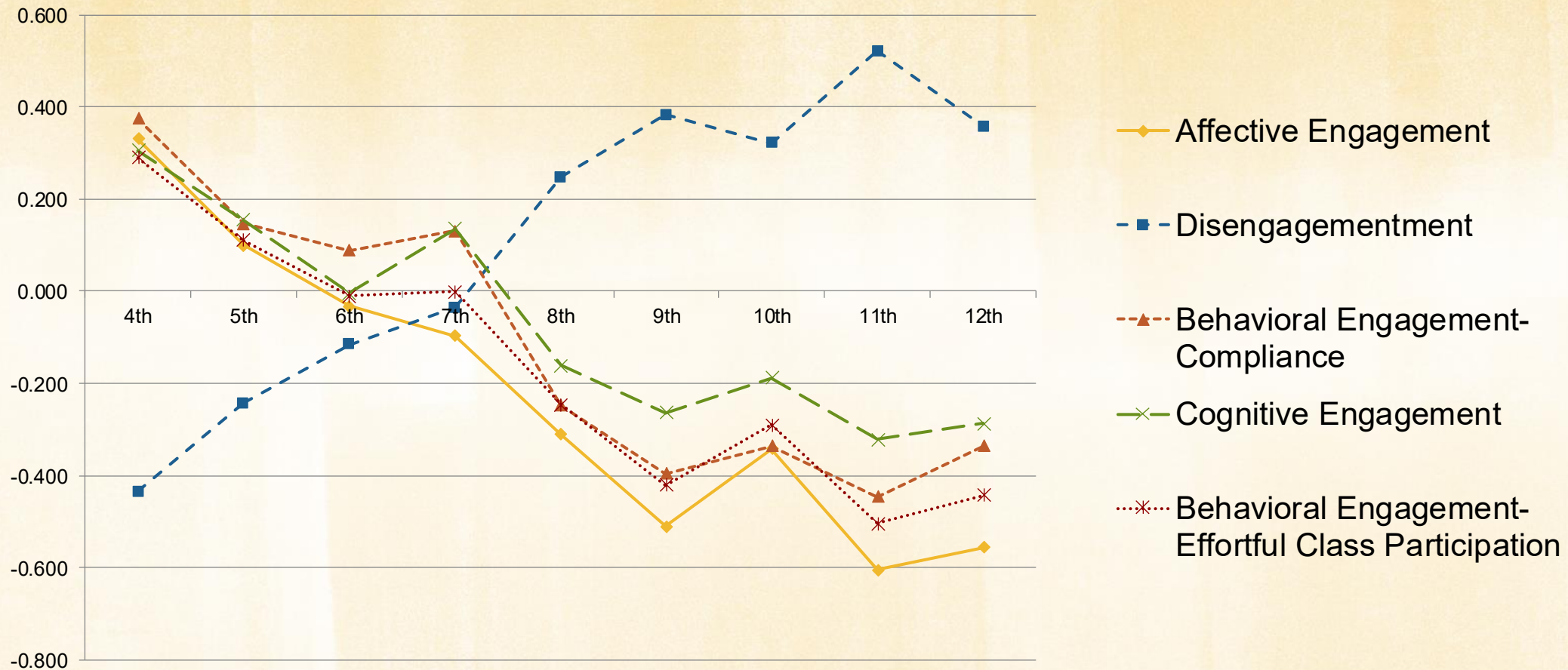
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Age Trend Across All Classes



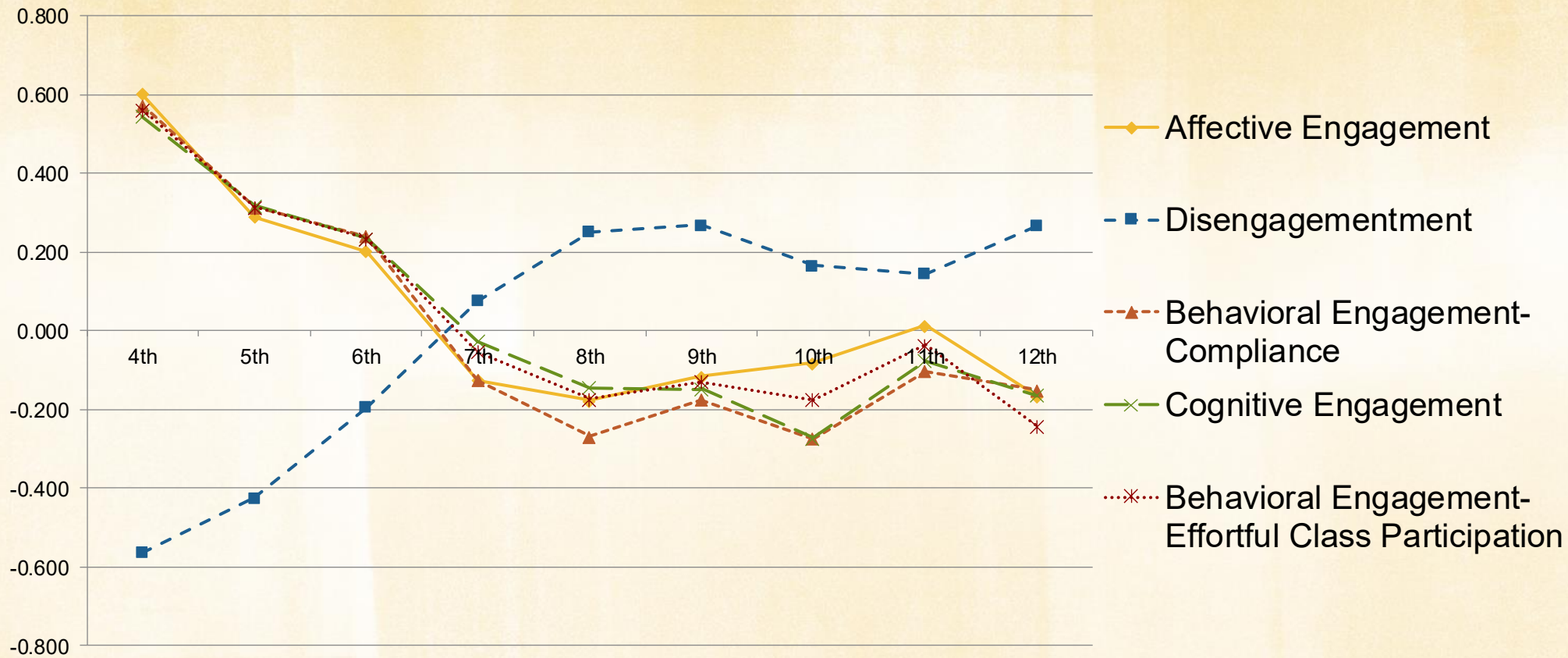
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Age Trend For Mathematics



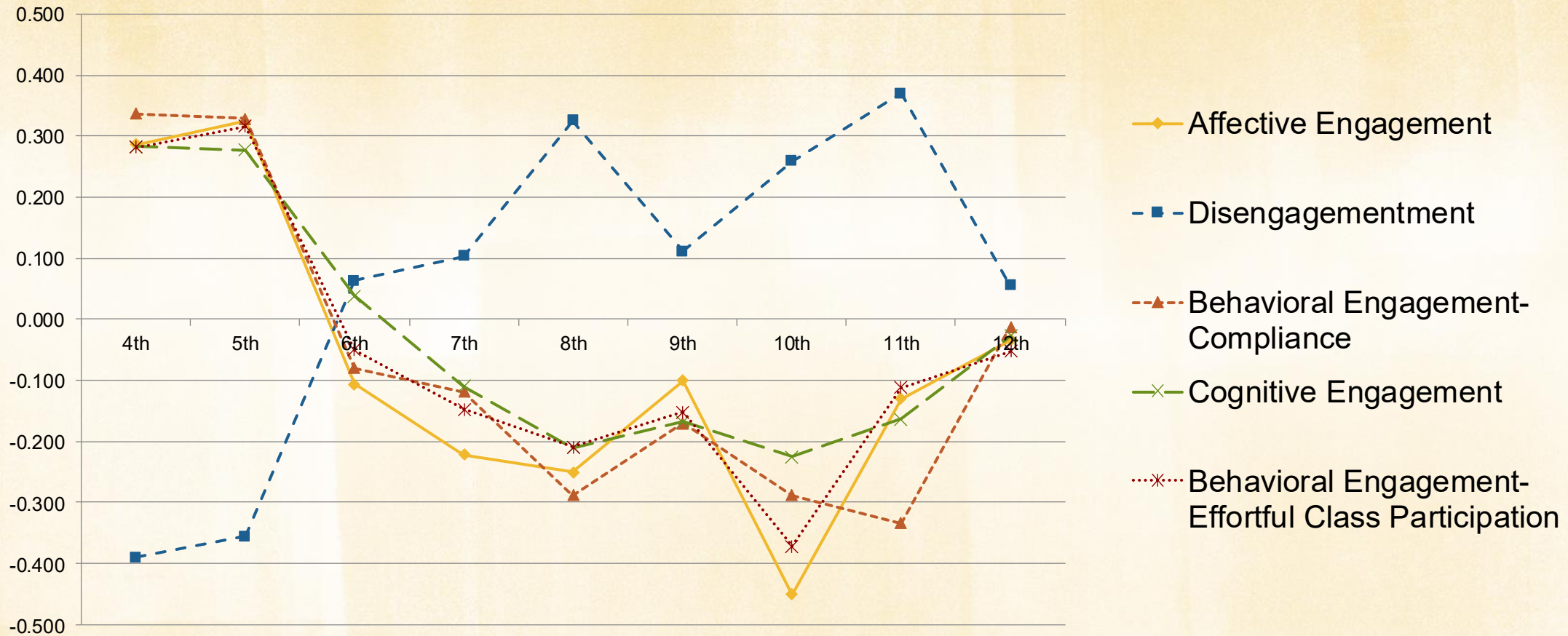
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Age Trend For Science



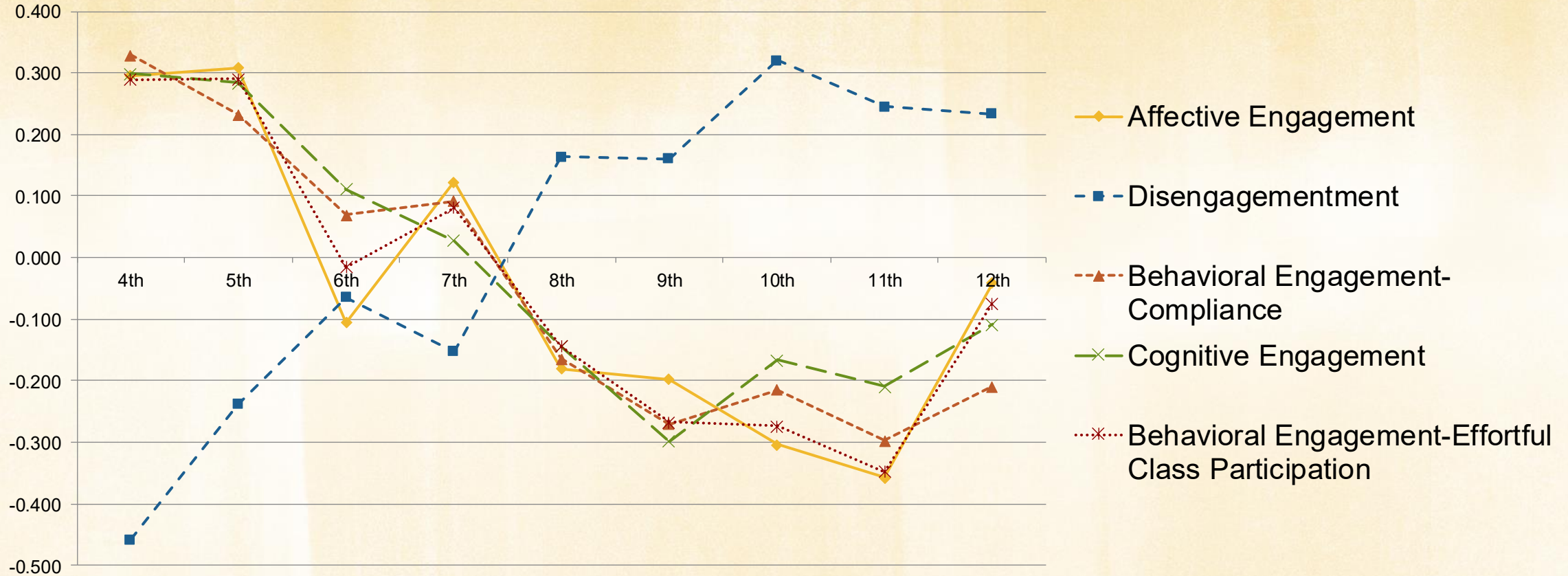
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Age Trend For Language Arts



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Age Trend For Social Studies



Age Trends in Classroom Engagement

- Vary by subject
 - For example, engagement in math steadily declined, with no uptick in later grades.
- Vary by type
 - For example, in 10th grade language arts, *Affective Engagement* and *Behavioral Engagement: Effortful Class Participation* drop precipitously compared to *Cognitive Engagement*, suggesting there might be a problem with the curriculum at that grade

Age Trends in Classroom Engagement

- The distressing decline in engagement across childhood may improve as youth approach young adulthood.
- This improvement could represent real engagement, or it could be driven by 11th and 12th grade students taking more electives that they like, and more alienated students being absent due to dropping out.
- In order to understand age trends in engagement, one needs to specify the content area and the domain of engagement.

CEI Has Been Used by Others

- Received email requests from 70+ researchers
- Researchers from Brazil, Canada, China, Malaysia, Spain, Turkey, United Kingdom, and United States
- Been cited 370+ times
- “What Teens Want From Their Schools”, a report put together by researchers from the Thomas B. Fordham Institute. A pdf copy of the report can be downloaded at:
<https://fordhaminstitute.org/national/research/what-teens-want-their-schools-national-survey-high-school-student-engagement>